

4.0 Maintenance



WARNING! ALL PERSONNEL WORKING ON OR AROUND THIS MACHINE MUST UNDERSTAND PROPER FUNCTION AND LOCATION OF SAFETY DEVICES. PROPER LOCKOUT/TAGOUT PROCEDURES SHOULD BE FOLLOWED AT ALL TIMES WHILE WORKING ON MACHINE

4.1 Scheduled Maintenance

A. Daily.

1. Check all safeties, switches and E-stops for proper function.
2. Verify that all electrical enclosures, leads are closed and sealed.
3. Check Pinch Block for cracks or chips. Flip over to use other edge or replace if necessary.

B. Weekly (all above +):

1. Check Paddle Wheel for unusual wear, chips, and cracks. If excessive wear is observed and/or skinner performance is deteriorating, replace.
2. Check all bearings for smooth movement.
3. Apply grease to all bearings.
4. Check for any damage or unusual or excessive wear, replace, repair parts as needed.
5. Check Upper Wearstrip assembly belt tension. Belting should not sag or drag across PaddleWheel/Pinch Block. Tighten when necessary.
6. Check all roller chain tension. Tighten when necessary.
7. Check all conveyor belt tensions. Tighten when necessary. Replace any broken or missing segments as needed.
8. Check all electrical enclosures, verify that they are sealed and that there is no standing water in any of them.
9. Check Water Filter and Filter Screen for debris and blockage.
10. Verify that Paddlewheel nozzles are not clogged or blocked. Blocked nozzles can result in improper skin ejection and premature Paddlewheel wear.

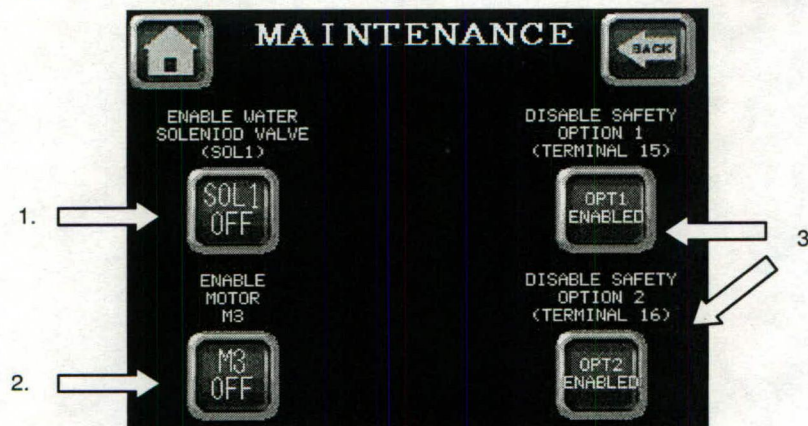
C. Monthly (all above +):

1. Check all fasteners and set screws and verify that they are snug. Tighten as necessary
2. Check Chain and sprocket wear (replace as needed).

D. Six months (all above)

4.2 Maintenance Screen

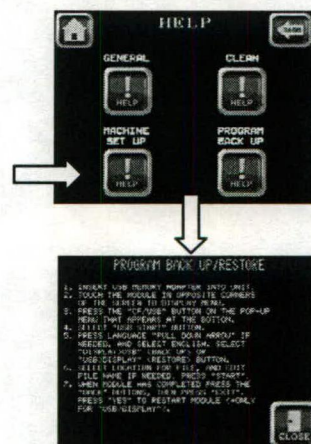
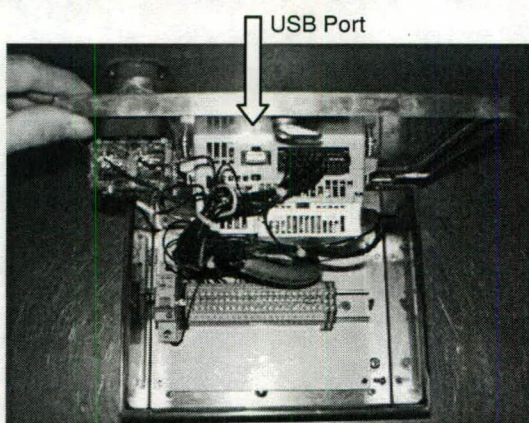
A. The maintenance screen can be accessed by the lower right hand button on the main menu. There you can turn off and on the optional controls inputs and outputs.



1. If using an optional solenoid valve to automatically shut off units water supply when not in operation it must be enabled through this screen after installation.
2. If using a third motor on the unit such as a separate drive for the exit conveyor.
3. If any additional safeties or switches are to be added they can be enabled with the buttons on the right hand side a screen. (Up to two additional devices can be used. Option 1 and 2 are bypassed in clean mode.)

4.3 Program Back Up/ Restore

A. The CSK24 unit's programming can be backed up and restored by up/downloading to and from a USB drive. The USB port can be found by opening the enclosure which houses the touch screen display, port is on the bottom of the HMI touch screen unit.



Instructions for this can also be found through the main help screen under Backup.

1. Insert USB memory adapter into unit.
2. Touch the Module in opposite corners of the screen to display menu.
3. Press the "CF/USB" button on the pop-up menu that appears at the bottom.
4. Select "USB START" button
5. Press the language "pull down arrow" if needed and select English. Select "DISPLAY>USB" (to backup) or "USB>DISPLAY" (to restore) button.
6. Select location for file, and edit file name if needed. Press "START".
7. When module has completed press the "BACK" buttons, then press "EXIT". Press "YES" to restart module (only for "USB>DISPLAY").

4.4 Troubleshooting

A. Skinner Will Not Start

1. Check Touch screen for any alarms that may be present. If there are any alarms present, lower left icon will read alarm as shown on previous page. Pressing this button will take you to the alarm screen and will explain the alarm. The alarms displayed are active. As guards are replaced and safeties reset, the alarms will reset automatically.
2. Check all power supplies, relays and disconnects.

B. Product Jams

1. Check all clearances between Star Wheel and Pinch Block for blockage.
2. Check height of upper assembly to pinch block to ensure upper assembly is pulling product through machine. Upper assembly may require vertical adjustment.

C. Removing Excess Meat with Skin

1. Adjust clearance between Star Wheel and Pinch Block. Adjust Pinch Block Take-ups evenly on both sides so that the Pinch Block is gently pushed against the Star Wheel. You should hear a slight chatter.
2. Try a less aggressive Paddlewheel. Paddlewheels are made with varying number of flights ranging from 12 points to 30 points per wheel. The lower the number the more aggressive the Paddlewheel.

D. Skin wrapping on Paddlewheel.

1. Check for proper Counter Roller, Paddle Wheel gap. A 0.015 to 0.030" gap must be maintained. If gap is over 0.030 skin may wrap on the Paddle Wheel instead of being ejected. Skin wrap can result in damage to unit. Reset gap following instructions in section "4.6 Counter Roller Adjustments."
2. Check Paddlewheel nozzle alignment. See section "4.5 Star Wheel Replacement/Nozzle Alignment" of this manual.

E. Skin remaining on some product.

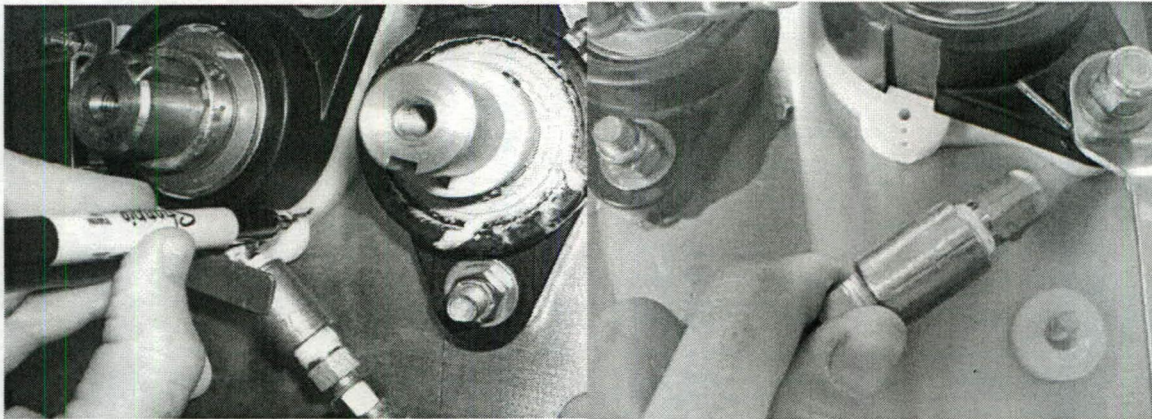
1. Excessive Paddle Wheel wear. Replace Paddle Wheel.
2. Uneven Paddle Wheel wear: Pinch block bolts tightened with wrenches and or tightened unevenly. Replace Paddle Wheel.
3. Product loading issues.
 - a) Product must be loaded skin down.
 - b) Product must be flat on loading belt.
 - c) Product should go into the skinner keel first.
 - d) Product must not be frozen.
 - e) Product must not overlap other product.
4. Chipped pinch block
 - a) If the upper assemblies belt becomes loose it can drag on the pinch block. This will cause a sharp or jagged edge that will rip skin.

For questions contact Prime Equipment Group at 614-253-8590
www.primeequipmentgroup.com

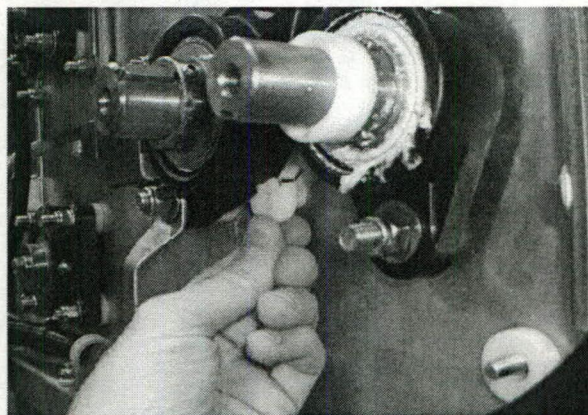
4.5 Star Wheel Replacement / Nozzle Alignment

A. Replace Star Wheel

1. Disconnect power to the skinner, lock out the power switch, and follow all other safety procedures at your facility.
2. Open hood guard and remove both chain covers. With nozzle removed from holder you should see two marks below the hole for nozzle on the spray nozzle holder. This is an alignment aid. Default position is with the two marks aligned vertically. The hole in the nozzle holder is drilled at a slight angle so when the holder is turned it adjusts the angle of the water stream. If the holder had to be moved from default to fine tune the direction of spray, mark the positions of the Nozzle Holders located at each end and below the Star Wheel as shown in the image below.



3. Remove Chain, Sprockets, loosen Star Wheel bearings, pull out Spray Nozzle and remove Paddle Wheel Inserts from Side Panel. You should now be able to pull the Star Wheel out the Idle side through the hole in the Side Panel.
4. Install new Star Wheel centered in between the Side Panels.
5. Check position of Nozzle Holder. Realign as needed. Nozzle Holders are held in place by flanged bearings. See image below

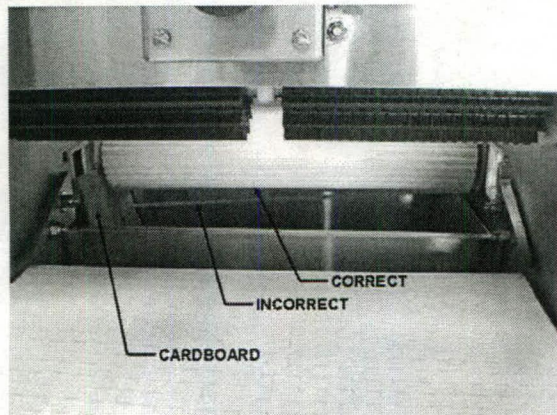


4.5 Star Wheel Replacement / Nozzle Alignment

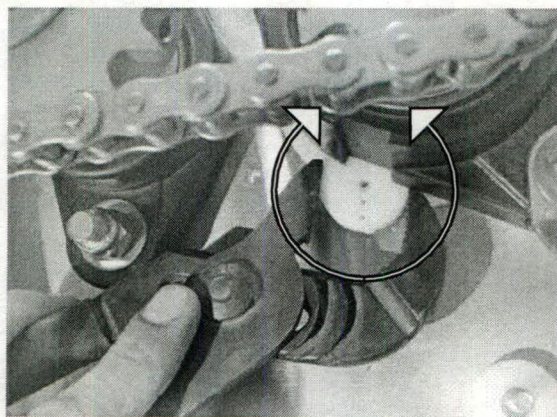
B. Nozzle Adjustment

If skin and fat are not being removed from the Star Wheel, there are three key areas to check: high pressure water supply, clogged nozzles, and misaligned nozzles. If the nozzles get out of alignment, please read through the following before adjusting.

1. Disconnect power to the skinner, lock out the power switch, and follow all other safety procedures at your facility.
2. Raise upper assembly(s) and hold in place with spring pins.
3. With Pinch Block and Exit Slide removed block one nozzle with a piece of for example cardboard. See image below.



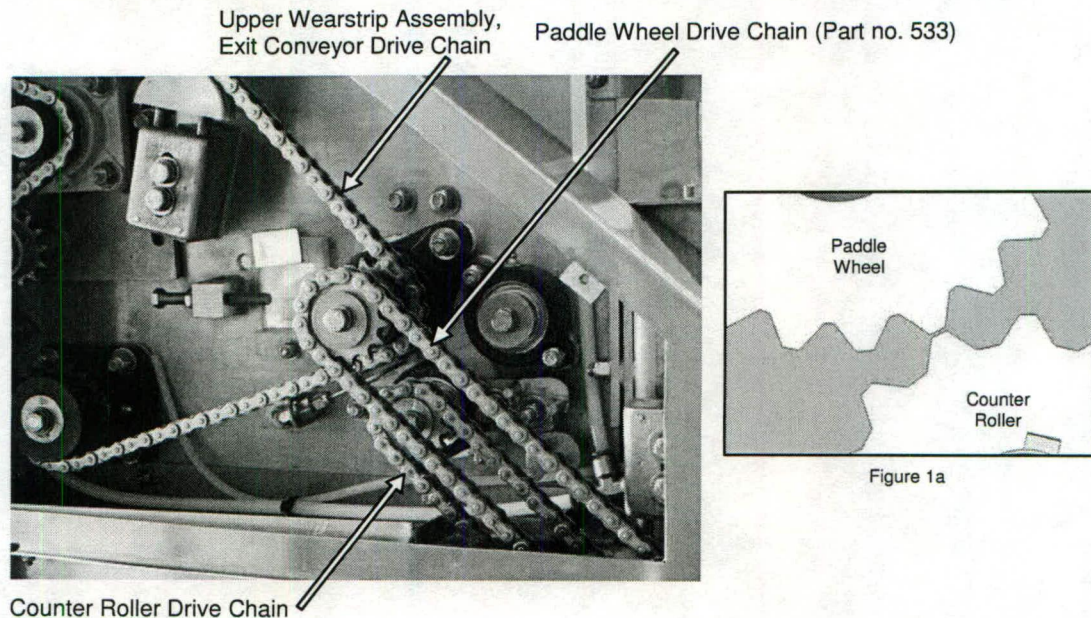
4. Turn on water and note water stream from the uncovered nozzle. Star Wheel may need to be moved slightly by hand to uncover water stream..
5. The water stream should be pointed toward and shooting between the teeth of the Star Wheel. See image above.
6. If adjustment is needed, loosen bearing and turn Nozzle Holder until the water stream is in the correct position. Turning the holder clockwise will aim the stream to the right and counter-clockwise will aim the stream to the left. See image below.



7. Repeat for other side.

4.6 Counter Roller Adjustments

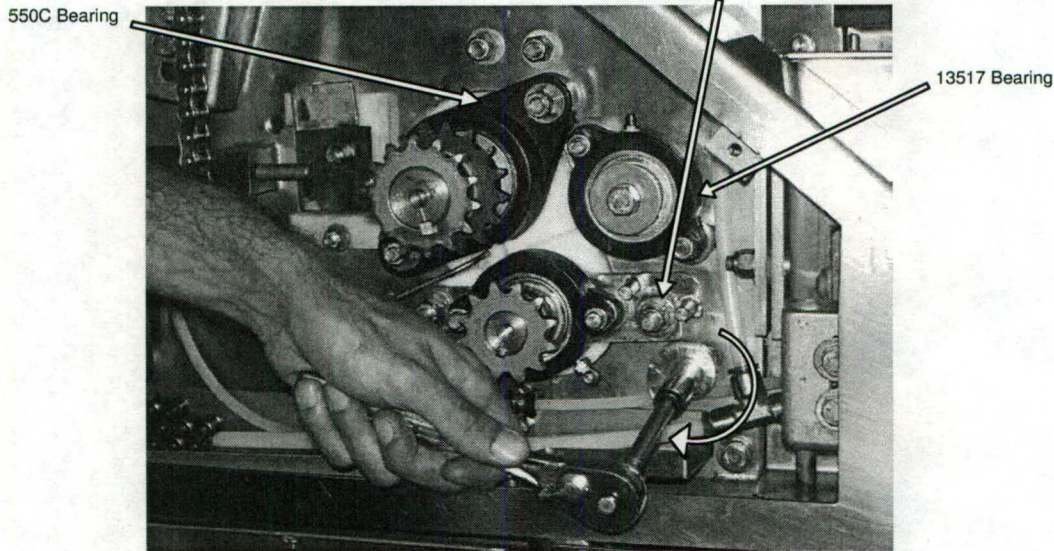
- A. The Counter Roller Removes Skin from the Paddle Wheel without the aid of high pressure water nozzles, preventing skin from wrapping around the Paddle Wheel. Paddle Wheel water nozzles are still used to keep it lubricated and to remove any particulate build up that may affect skinning performance but these nozzles operate at house pressure and do not require a pump. Skin wrap on the Paddle Wheel will prevent the skinner unit from operating properly and may lead to damage of the unit. For proper operation of the Counter Roller a 0.015 to 0.03" gap must be maintained between the Counter Roller and the Paddle Wheel. Paddle Wheels wear over time and can still function after losing 0.25" off of its original diameter of 3.5". The Counter Roller gap between the Paddle Wheel must be periodically checked and maintained. The Counter Roller is adjustable and can be moved closer to the Paddle Wheel as it wears away.
- B. Skin wrap on the Paddle Wheel may be an indication that the gap between it and the Counter Roller is too great. Particulate build up on the Paddle wheel may be a sign of clogged Paddle Wheel Nozzles, do not run the unit without Paddle Wheel Nozzles flowing freely. The Counter Roller is mounted on a pivoting plated which can be adjusted with the aid of cams. To Check and maintain the Counter Roller gap follow the steps below.
1. Shut off Skinner unit, Follow all Lockout/Tagout procedures.
 2. Open Skinner Lid, remove Rear Slide and Pinch Block. Remove Paddle Wheel and Counter Roller drive chains as well as the drive chain on the Paddle Wheel that powers the Upper Wearstrip assembly and or exit conveyor. Break Feed Conveyor belting so that the Paddle Wheel may spin freely, Feed Conveyor drive cover (on opposite side of unit) and drive chain should not have to be removed.



3. Rotate Paddle Wheel and Counter roller until they are flight to flight to make sure that the correct distance is being measured (See Fig. 1a) Using shim stock measure and verify the Counter Roller gap. Spin both Paddle Wheel and Counter Roller while shim is in place to make sure that both are flight to flight. If gap is in-between 0.015 and 0.3" no adjustment should be necessary. Do not attempt to set the gap closer than 0.015", this may lead to the Paddle Wheel and Counter Roller rubbing against each other. Never let the Counter Roller and Paddle Wheel to come in contact while in operation, this will lead to premature wear and damage to the Skinner unit.

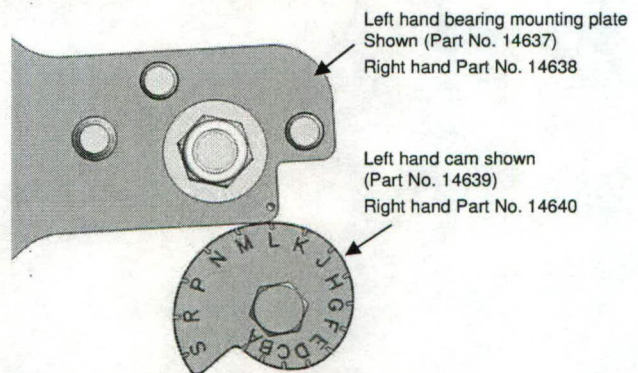
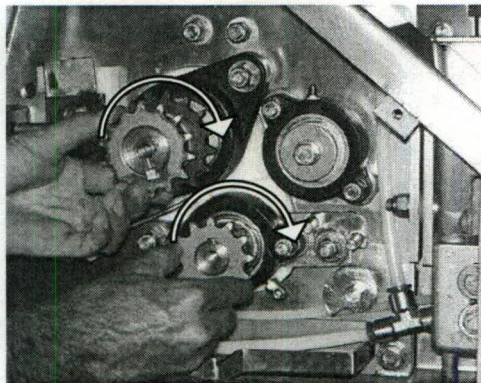
4. If Counter Roller gap is over 0.03" adjustment may be needed and continue below.
5. Loosen Counter Roller Bearing Mounting Plate by using the Nut shown Below. This will loosen the Counter Roller bearing mounting plates on both sides of the machine. Feed conveyor drive cover will not have to be removed.

Loosen this nut so Counter Roller (Part No. 14635) is free to adjust (Tension Rod Part No. 14939)



Turning cam to raise Counter Roller

6. With the Paddle Wheel and the Counter Roller Flight to Flight spin the cam to raise the Counter Roller until it begins to touch the Paddle Wheel. Verify that the Flights on the Counter Roller and Paddle Wheel are not intermeshed by spinning both the Paddle Wheel and Counter Roller at the same time. Both should be free to spin and should be just brushing each other, not intermeshed and locked up, if so, reset.



7. Once that it is confirmed that the outer Diameters of the Paddle Wheel and Counter Roller are tangent, back the counter roller away from the Paddle Wheel. Before turning the cam look to see which letter is facing the Counter Roller bearing mounting plate and the back the cam down one letter. Turning the cam one letter will increase or decrease the roller gap by 0.015" +/- 0.005". Once gap is set spin both Paddle Wheel and Counter Roller again to verify that they are not rubbing or brushing against each other, if so back down one and a half more letters and recheck.
8. Once Gap is set, tighten nut on the Counter Roller Tension Rod, locking both bearing plates on each side of unit into place, tighten all fasteners and reinstall drive chains.